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Indian Sign Language Recognition System for Dynamic Signs

Arun Singh
Student, Integrated B. Tech M. Tech, Computer Science and Engineering, Lovely Professional University
Phagwara, India
arunmandiarun2001@gmail.com

Ankita Wadhawan
Department of Computer Science and Engineering
Lovely Professional University
Jalandhar, Phagwara, Punjab, India
Ankita.23891@lpu.co.in

Manik Rakhra
Department of Computer Science and Engineering.
Lovely Professional University
Jalandhar, Phagwara, Punjab,India
Rakhramanik786@gmail.com

Usha Mittal
Department of Computer Science and Engineering
Lovely Professional University
Jalandhar, Phagwara, Punjab, India
usha.20339@lpu.co.in

Ahmed Al Ahdal
Department of Computer Science and Engineering
Lovely Professional University
Jalandhar, Phagwara, Punjab, India
ahmed.alahdal21@gmail.com

Shambhu Kumar Jha
AIIT,
Amity University
Uttar Pradesh Noida
India skjha2@amity.edu

***Abstract*—Sign language is a means of communication utilising manual gestures (movement of hands and wrists) and non-manual gestures (expressions of face and body language). There are many different sign languages in the world, each with its own collection of words and signs. This study focuses on the implementation of Indian Sign Language Recognition System (ISLRS) which help deaf people to communicate with other persons. In this paper, the model based on Sign Language Recognition (SLR) of dynamic signs using Convolutional Neural Network (CNN) is proposed. The proposed model has been trained and tested on video clips of dynamic signs and achieved the training accuracy of 70%. This study should serve as a road map for users to use in deciding which model to implement and laying a foundation for future research and enhancing model accuracy, allowing the sign language community to communicate and share their ideas more effectively. This work is also used to overcome the educational gap between hearing impaired persons.**

Keywords— Sign language Recognition, Classification, Feature Extraction, Machine Learning, Deep Learning, Long Artificial Neural Network (ANN), Support Vector Machine (SVM).

I. INTRODUCTION

Language is crucial in communication and interaction between two or more people, however deaf individuals rely on Sign language because they do not have the gift of speech (SL). The deaf and dumb community rely heavily on sign language to communicate with other persons. Instead of lip-reading or reading facial emotions, sign language is a good choice for them to consider. Additionally, they use non-verbal cues such as facial expressions to better grasp their emotions, making it easier for them to carry forward their feelings.

Various steps used to carry out the process of recognition and making the system understandable by the computer are data pre-processing, segmentation, feature extraction, pattern matching, class recognition, computer vision, natural language processing (NLP), and selecting a perfect model. The models are designed in such a manner so that the recognition system can be deployed in real-time for improved human to human interaction and human-computer interaction which also reduces the educational The models are designed in such a manner so that the recognition system can be deployed in real-time for improved human to human interaction and human-computer interaction

which also reduces the educational gap among hearing impaired persons. It also makes the hearing-impaired people more independent and self-reliant in public spaces like buses, markets, hotels, offices etc.

The main challenge faced by the researchers during this research is collecting data from different people from different backgrounds. This is the most time-consuming process in the whole work because without data you cannot train and test your model. Second, selecting whether to go for a deep learning approach or a machine learning approach. Then choose a model that would be implemented and if the accuracy comes out to be very low then go for another model or improve its accuracy.

II. TYPES OF SIGN LANGUAGE AND THEIR CLASSIFICATION

Sign language was invented in the early 1970s with different signs for alphabets, numbers, words, and symbols. The signs are conveyed using hands, wrists, and non-manual expressions. The orientation of the hands and wrists specify different signs and letters; sign language gestures are classified into 3 categories i.e., signs with one hand, non-manual, signs with both the hands; these are further categorized into static and dynamic signs as shown in Figure 1.

Single-handed signs are associated with a single dominant hand and this sign can be static or a dynamic sign; static signs are those signs that does not involve any motion of the hand, on the other hand the dynamic signs are the signs that involve the motion of hands or other body parts.

Double-handed signs are also divided into static and dynamic signs which are further subdivided into T1 and T0 i.e., Type1 and Type0. T1 signs use the dominant hand more than the non-dominant hand and T0 is vice versa. Non-manual signs are those which are expressed without using hands instead of using facial expressions, body postures, or mouth gestures.

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Non-manual signs are those which are expressed without using hands instead of using facial expressions, body postures, or mouth gestures.

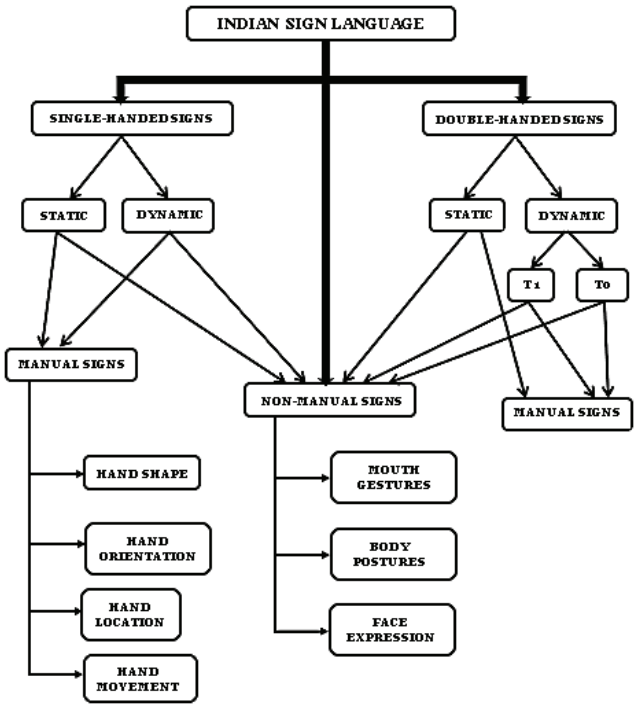


Fig. 1. Indian Sign Language hierarchy.

III. LITERATURE REVIEW AND PLANNING

Figure 2 depicts the many criteria used in the fact comparison. Our plan for the literature review is to compare machine learning and deep learning techniques using isolated signs (single sign gestures) versus continuous signs (a sentence using ISL), the number of signs, the total dataset size, feature extraction technique to extract hidden characteristics, classification technique to classify the data in the respective sign, and the accuracy of the model they used. The main objective of this section is to go through latest research articles on Indian Sign Language (ISL) recognition systems, thus we only considered papers with the keyword ISL from 2017 to 2021.

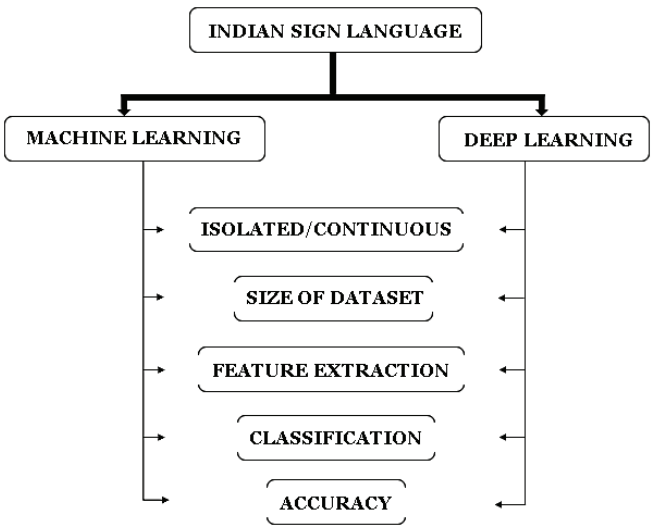


Fig. 2. Reviewing classification

IV. MACHINE LEARNING TECHNIQUES

In 2017, Ekbote and Mahasweta Joshi developed Indian sign language categorization system using support vector machines and neural networks. The features in this research get extracted by using different techniques like Scale.

Invariant Feature Transform (SIFT), and Histogram of Oriented Gradients (HOG) and shape descriptors. The experiments were performed on 10 isolated single-handed signs (0 to 9 numerals) and classified using ANN and SVM. The experimental findings revealed that both SVM and ANN had a 93% accuracy [1].

In 2017, Jadav and Rokade developed an ISLRS using 26 one handed and double-handed alphabetical signs and extracted characteristics using Distance Transformation, Discrete Fourier Transform, and the central probability distribution property, then categorised them using ANN and SVM. The results of the experiments revealed that ANN outperformed SVM [2].

In 2017, Hore et al. developed an optimized neural network based Sign Language Recognition System (SLRS). They gathered 22 one-handed and double-handed gestures. They used a Genetic algorithm technique and Euclidean distance for extracting the characteristics, and K-nearest neighbour (KNN) and Euclidean distance for classification. The NN- PSO beat the other techniques in the experiments, with 99.96 accuracies, precision of 99.98, 98.29 recall, 99.63 F-Measure, and kappa statistics to be 0.9956 [3].

Rao and Kishore proposed selfie video-based continuous ISLRS in 2018. They have collected and used 50 continuous single-handed signs. They have extracted the features using Sobel adaptive threshold, Morphological differencing and for classification, the models used are Minimum Distance, Artificial Neural Network. The results present that ANN outperformed with 90% accuracy [4].

Shenoy et al. (2018) proposed a "Real-time Indian Sign Language (ISL) Recognition". They have isolated and continuous 33 hand postures and 12 gestures employing single-hand and double-hand. Grid-based Feature Extraction, Feature Vector, and HOG are the feature extraction techniques employed, while the classification model used is KNN, Hidden Markov Model (HMM). The results reveal a 99.7% accuracy for postures with static hand and a 97.23% accuracy for the recognition gestures [7].

Mariappan and Gomathi developed a SLRS to recognize signs in real-time in 2019.They have collected 80-word signs and 50 continuous sentences for their dataset using both single and double hands for feature extraction they have used orientation histogram, eigenvector-based technique, and classification technique used is eigenvalue weighted euclidean distance technique. The classification accuracy rate thus obtained is 86% [8].

Athira et al. presented a signer independent SLRS with elimination of co-articulation from live videos in 2019. They have collected 46 isolated and used SVM. Experimental results show that SVM has produced an accuracy of 92.88% [10].

In 2020, Raghuveera et al. proposed a depth-based ISLRS using Microsoft Kinect. They have collected 140 unique isolated and continuous gestures using single-hand and double-hand. Feature extraction has been done by

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Speeded Up Robust Features, HOG, and Local Binary Patterns; and the collected signs were categorized by using SVM. Experimental results state that the model produced an accuracy up to 71.85% [12].

Badhe and Kulkarni proposed an artificial neural network based ISLRS using handcrafted features' in 2020. They have collected 46 different continuous signs using single and double hand; for feature extraction the techniques used are HOG, Principal Curvature Based Region (PCBR), and Wavelet Packet Decomposition (WPD), Principal Component Analysis (PCA), or Scale Invariant Feature Transform (SIFT). The classification technique used is Artificial Neural Network (ANN) and after rigorous work secured training accuracy 98%, validation accuracy 63%[27].

TABLE I. MACHINE LEARNING TECHNIQUES

Author	Isolated/ Continuous	Single/ DoubleHanded	Signs	Total DatasetSize	Feature Extraction Technique	lassification Technique	Accuracy
Ekbote and Mahasweta(2017)	Isolated	Singlehanded	10 Signs (0 to 9 numerals)	1000 images,	SIFT and HOG	Artificial Neural Networks (ANN) and Support VectorMachine (SVM)	93%
Jadav andRokade (2017)	Isolated	Both	;ns of26 Alphabets	260 images are	distance transformation, Discrete Fourier Transform, Probability distribution property that is central moments	Artificial Neural Network and SVM	94.37% (ANN), 92.12%(SVM)
Hore *et al.* (2017)	Isolated	Both	22 ISL gestures	220 images	Genetic algorithmand Euclidean distance	. K-nearest neighbour and Euclidean distance	99.96 accuracy, 99.98 precision, 98.29 recall, 99.63 F- Measure, and0.9956 Kappa Statistic
Rao and Kishore (2018)	Continuous	Single-handed	50 Signs	2500 samples	Morphological differencing andsobel threshold	Minimum Distance, ArtificialNeural Network	85.58% (MDC) and 90% (ANN)
Shenoy *et al.* (2018)	Both	Both	33 hand poses and 12 gestures	24,624 images	Grid-based Feature Extraction, Feature Vector, Histogram of Oriented Gradients (HOG).	KNN and HMM	99.7% (static hand postures) , and 97.23% (gestures)
Mariappan and Gomathi(2019)	Continuous	Both	80 Word signs and 50 Sentence signs	800 (words) , 500 (sentence) = 1300.	Orientation Histogram, eigenvector-based technique	eigenvalue weighted Euclidean distance technique	86 %
Athira *et al.* (2019)	both	Both	46 ISL signs	900 static images and700 videos	HOG, HOEF and Zernike moments	Support Vector Machine (SVM)	92.88%
Raghuveera *etal.* (2020)	Both	Both	140 unique gestures	4600 images	HOG and LocalBinary Patterns.	Support Vector Machine (SVM)	accuracy up to 71.85%
Badhe and Kulkarni (2020)	Continuous	Both	46 different signs	500 videos of 10 different sign gestures235 images of 36 signs	HOG, PCBR andWPD, PCA, SIFT	Artificial Neural Network	Trainingaccuracy 98%, Validation accuracy 63%

V. DEEP LEARNING-BASED TECHNIQUES

Rao et al. (2018) proposed a "Deep Convolutional Neural Networks for Sign Language Recognition". They employed a Convolution Neural Network (CNN) with four convolutional layers, two stochastic pooling layers, and Softmax regression to classify 200 continuous sign images taken with single and double hands. It has a recognition rate of 92.88 percent, according to experimental results [5]. "Indian Sign Language Numeral Recognition Using Region of Interest Convolutional Neural Network" was proposed by Sajanraj T D and Beena M V in 2018. They gathered 36 isolated single and double-handed signs, using the CNN, HSV, and colour space with Contrast Limited Adaptive Histogram Equalization as the classification model (CLACHE). In low light conditions, the experiment produced 99.56 % accuracy and 97.26 % accuracy [6].

In 2019, Ankita Wadhawan and Parteek Kumar suggested a "Deep learning-based static sign language identification system." They employed single and double hands to gather 100 static continuous signs, and the model they used was a CNN with max-pooling. On coloured and grayscale images, the experiment yielded the highest training accuracy of 99.72% and 99.90 %, respectively [9].

Neel Kamal Bhagat, Vishnusai Y, Rathna G N proposed “Indian Sign Language Gesture Recognition using Image Processing and Deep Learning” in 2019. They collected 36 isolated static gestures, 10 dynamic signs using single and double-hands the classification model used is CNN, Long Short-Term Memory (LSTM), Connectionist temporal

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classification, and Handcrafted feature techniques like Scale Invariant Feature Transform (SIFT), s Gradient Hough Transform (GHT). The experiment produced an accuracy of 97.71% [11].

In 2021, R. Dhiman, G. Joshi, and C. Rama Krishna proposed "A deep learning strategy for classification of Indian sign language gestures with varied backgrounds." They employed single and double hands to gather 15 isolated indicators; the classification model was CNN models softmax layer with different optimization techniques, and activation functions. The accuracy for NUS was 95.95 percent and 97.22 percent, respectively [14] dataset-I, NUS dataset-II, and the mixed dataset is 100%, Adithya Venugopalan, Rajesh Reghunadhan proposed "Applying deep neural networks for the automatic recognition of sign language words: A communication aid to deaf agriculturists" in 2021. They have collected 9 continuous double-handed gestures; the classification model used is a CNN, GoogleNet, and bidirectional LSTM (BiLSTM) sequence network. The assessment produced an average classification accuracy of 76.21% [15].

TABLE II. DEEPLEARNING-BASED TECHNIQUE

Author	Isolated/ Continuous	Single/ Double Handed	Signs	Total DatasetSize	Feature Extraction Technique	ClassificationTechnique	Accuracy
Rao *et al.* (2018)	Continuous	Both	200 signs	120000 sign images	Softmax regression	CNN	92.88%
Sajanraj andBeena (2018)	Isolated	Both	36 signs	3000 images	CLACHE (Contrast Limited Adaptive Histogram Equalization)	CNN, HSV colorspace	99.56% and 97.26% (low light condition)
Wadhawan and Kumar (2019)	Continuous	Both	100 static signs	total 35,000 sign images	max-pooling	CNN	99.90%
Bhagat *et al.* (2019)	Isolated	Both	36 static gestures, 10 dynamic signs	45,000 images based out of the depth camera, RGBcamera, images without background segmentation	Handcrafted feature techniques likeSIFT, GradientHough Transform (GHT)	CNN, LSTM, Connectionist temporal classification	97.71%
Dhiman *et al.* (2021)	Isolated	Both	15 signs	2240 images	Convolution Neural Network (CNN) model softmax layer	CNN	100% (NUS dataset-I), 95.95% (NUS dataset-II), and 97.22% (mixed dataset)
Venugopalan and Reghunadhan (2021)	Continuous	Double Handed	9 hand gesture classes	900 video samples of nine hand gesture classes	CNN, GoogleNet	bidirectional LSTM (BiLSTM)	76.21%

VI. PROPOSED METHODOLOGY

A. Data Acquisition

The initial stage in starting the research is data collection, which is the most significant aspect of the experiment. We have acquired dynamic video data of 50 signs as shown in Figure 3.



Fig. 3. people performing the signs

The signs[32,33] we recorded using mobile phones from 18 different people and each sign performed 50 times i.e., 18x50x50 more than 25,000 signs were collected and sorted into its respective folders as shown in Figure 4.

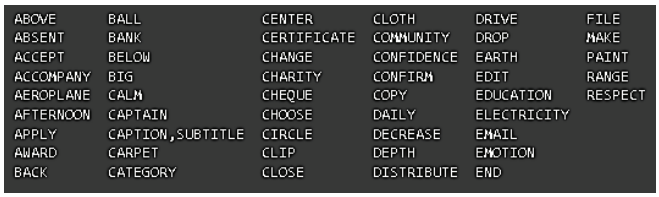


Fig. 4. Signs we collected

B. Work Flow And Model

Initially as discussed earlier we collect videos from 18 people and then we sort them in the respective folders. Then we pre- process and normalize the data. The system flow has been given in Figure 5.

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Fig. 5. System Flow

We have prepared a CNN Model picking up the best features from the above listed papers; the procedure we followed is generate the dataset and partition it into training and testing sets, and create a model with relu activator and softmax pooling layers and below are the further details of the model.

```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d (Conv2D)              (None, 58, 58, 64)        1792

conv2d_1 (Conv2D)            (None, 56, 56, 64)        36928

batch_normalization (BatchN  (None, 56, 56, 64)        256
ormalization)

max_pooling2d (MaxPooling2D  (None, 28, 28, 64)        0
)

global_average_pooling2d (G  (None, 64)                0
lobalAveragePooling2D)

dense (Dense)                (None, 256)               16640

batch_normalization_1 (Batc  (None, 256)               1024
hNormalization)

dense_1 (Dense)              (None, 49)                12593

=================================================================
Total params: 69,233
Trainable params: 68,593
Non-trainable params: 640
_________________________________________________________________
Model Created Successfully!
```

Fig. 6. Created model

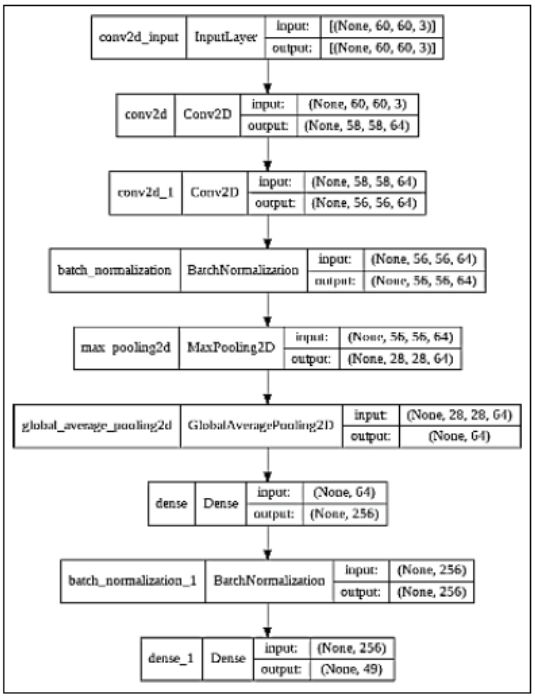


Fig. 7. Layers and features of the model

VII. Result Analysis

Analyzing the data of ISL in the recent times till 2021as shown in Figure 8 it is seen that in 2018 and 2019 the most papers were published and in 2021 the least papers were published From the data summary, it is seen that both isolated and continuous signs are both used equally over the years but both combined are not used much as shown in Figure 9.

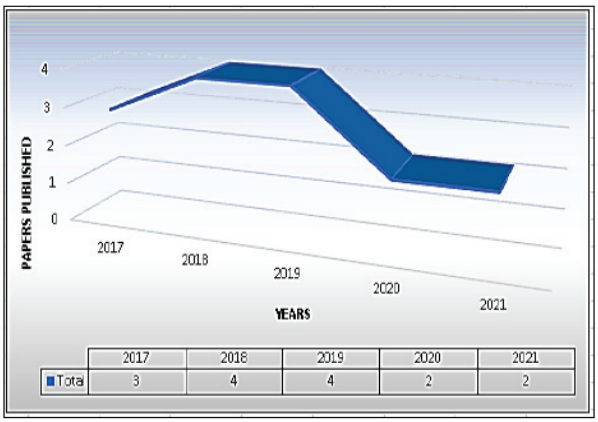


Fig. 8. Trendline of papers published over the years.

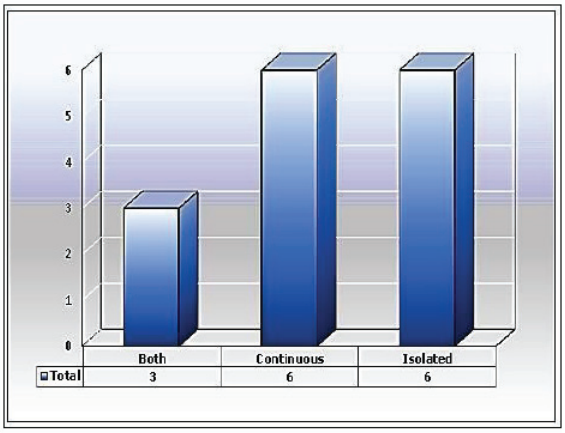


Fig. 9. Research work on isolated and continuous signs.

The datasets used mostly by people have both single-handed and double-handed signs in them having both static and dynamic gestures as shown in Figure 10.

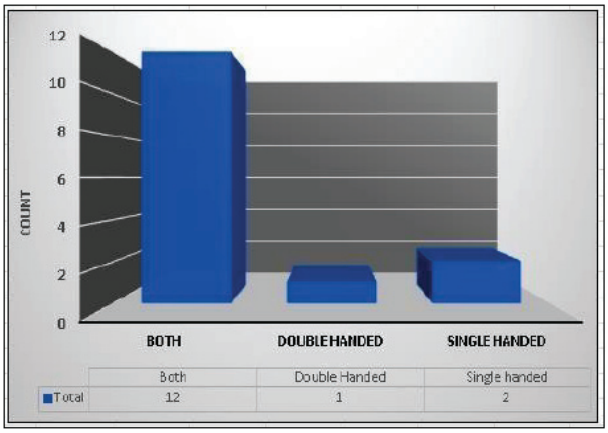


Fig. 10. Research Work for Single/Double Handed Signs

A pie-chart representation of all the classification techniques used by different peoples working on ISL recognition and it is seen that the CNN model is used the most in deep learning techniques the classification accuracy it's getting is more than 96% in the all above studies and ANN is used the most for machine learning techniques and the accuracy it is generating is more than 90%. So, this clearly shows that the CNN model is better to be used as shown in Figure 11.

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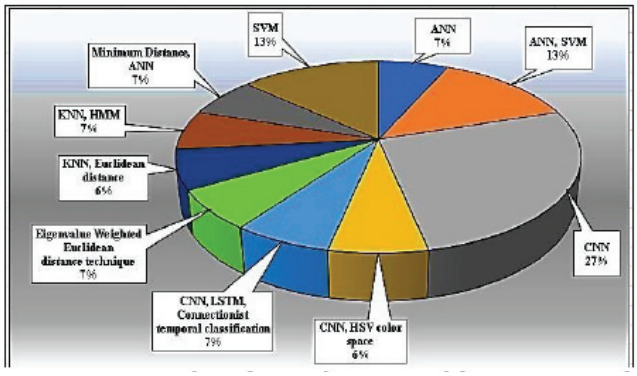


Fig. 11. Research work on techniques used for recognition of signs.

VIII. Conclusion And Future Scope

This paper analysed all 15 research articles on Indian sign language published in recent times till 2021, which offer a wide range of applications that will benefit the deaf community. This aims to summarise in Indian Sign Language based on isolated/continuous, single/double-handed motions, signs, total dataset size, feature extraction approach, classification technique, and accuracy. This article offers sufficient information to allow anyone to get a head start on their job without having to rewrite it; the following are the results:

- It has been observed that the most papers were published in the years 2018 and 2019, and that the number of papers released after that has declined, and that individuals have begun to research using deep-learning approaches more frequently than machine-learning techniques in the past.
- The majority of people have utilised both single and double-handed motions for their dataset, which was filmed using a camera or a mobile phone utilising both single and double hands isolated and continuous.
- According to the findings, deep-learning based approaches such as CNN outperformed other approaches in terms of feature extraction and classification with an accuracy of over 95%, which is exceptional, whereas other traditional machine-learning techniques produced less accuracy.

Using the model we constructed based on the features we found in articles, we were able to achieve a training accuracy of 70%. It's less because the videos were filmed in various lighting and background situations with smartphones of varying camera quality, and a couple of the clips were horizontal and vertical in nature, making feature extraction much more difficult, but it's a genuine situation. With eliminate all of the noise in the films, the model's accuracy and performance must be improved by the use of GPUs and better feature extraction techniques, as well as data standardisation. This research also help hearing impaired people to come forward get education and interact with other persons in the society. This also leads to the reduction in educational gap that exists among hearing impaired persons in the society.

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